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Beam delivery system

Abstract

A beam delivery system is provided that includes a beam delivery photonic integrated circuit. The beam delivery photonic integrated circuit includes one or more optical inputs; a plurality of waveguide outputs; and a plurality of beam paths. Each beam path connects one of the plurality of waveguide outputs to at least one of the optical inputs. The plurality of waveguide outputs are configured to emit a plurality of parallel beams. The beam delivery photonic integrated circuit is on a chip. The beam delivery system further includes a telecentric optical relay assembly. The telecentric optical relay assembly is configured to receive the plurality of parallel beams provided by the waveguide outputs and focus each received beam on a corresponding one of a plurality of positions of an atomic object confinement apparatus in a telecentric manner.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/316,354, filed May 10, 2021, which claims priority to and the benefit of U.S. Provisional Patent Application No. 63/037,482, filed Jun. 10, 2020, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(1) Various embodiments relate to a beam delivery system, and specifically to a beam delivery system for quantum computing applications with a beam delivery photonic integrated circuit.

BACKGROUND

(2) In various atomic systems, it is important to be able to deliver laser beams to dense one-or-more dimensional arrays of atomic systems at precision spacings and mode profiles. For example, trapped ion quantum computing uses laser beams to complete various functions within the trapped ion quantum computer. Such applications require that the lasers are delivered to the ion trap precisely and accurately in terms of position, spacings, mode profile, polarization, frequency, and phase.

BRIEF SUMMARY OF EXAMPLE EMBODIMENTS

(3) Example embodiments provide a beam delivery element for quantum computing applications.

(4) According to one aspect, a beam delivery photonic integrated circuit is provided. In an example embodiment, the beam delivery system comprises a beam delivery photonic integrated circuit, which comprises one or more optical inputs, a plurality of waveguide outputs, wherein each of the plurality of waveguide outputs are tapered, a plurality of beam paths, each beam path connecting one of the plurality of waveguide outputs to at least one of the one or more optical inputs, wherein the beam delivery photonic integrated circuit is on a chip, and an optical relay assembly, the optical relay assembly configured to receive beams provided by at least two of the plurality of waveguide outputs and focus each received beam on a corresponding one of a plurality of positions of an atomic object confinement apparatus.

(5) In another example embodiment, the one or more of the plurality of beam paths comprises an optical element configured to tune a beam for at least one of frequency, phase, or intensity.

(6) In another example embodiment, at least one of the plurality of waveguide outputs is directed to a photodetector.

(7) In another example embodiment, the plurality of beam paths are defined lithographically.

(8) In another example embodiment, the plurality of waveguide outputs are located on a polished edge of the beam delivery photonic integrated circuit.

(9) In another example embodiment, each of the waveguide outputs are on one side of the beam delivery photonic integrated circuit and are parallel to each other.

(10) In another example embodiment, each of the plurality of waveguide outputs are at most 150 micrometers from the next closest of the plurality of waveguide outputs.

(11) In another example embodiment, the optical input is butt-coupled to an optical fiber providing an input beam.

(12) In another example embodiment, a beam delivery system comprises a beam delivery photonic integrated circuit, which comprises a first waveguide layer and a second waveguide layer, the first waveguide layer, which comprises a first optical input, a plurality of first waveguide outputs, wherein the plurality of first waveguide outputs are tapered, a plurality of first beam paths, each first beam path connecting one of the plurality of first waveguide outputs to the first optical input, the second waveguide layer comprises a second optical input, a plurality of second waveguide outputs, wherein the plurality of second waveguide outputs are tapered, a plurality of second beam paths, each second beam path connecting one of the plurality of second waveguide outputs to the second optical input, wherein the beam delivery photonic integrated circuit is on a chip, an optical relay assembly, the optical relay assembly configured to receive beams provided by at least one of the plurality of first waveguide outputs and at least one of the plurality of second waveguide outputs, and the optical relay assembly further configured to focus each received beam on a corresponding plurality of positions of an atomic object confinement apparatus.

(13) In another example embodiment, one or more of the plurality of first waveguide outputs or of the plurality of second waveguide outputs is directed to a photodetector.

(14) In another example embodiment, one or more of the plurality of first beam paths or the plurality of second beam paths comprises an optical element configured to tune a beam for at least one of frequency, phase, or intensity.

(15) In another example embodiment, the plurality of first waveguide outputs and the plurality of

second waveguide outputs are located on a polished edge of the beam delivery photonic integrated circuit.

(16) In another example embodiment, each of the plurality of first waveguide outputs and the plurality of second waveguide outputs are on one side of the beam delivery photonic integrated circuit and are parallel to each other.

(17) In another example embodiment, each of the plurality of first waveguide outputs are at most 150 micrometers from the next closest of the plurality of waveguide outputs.

(18) In another example embodiment, at least one of the first waveguide layer or the second waveguide layer comprises a dielectric material.

(19) In another example embodiment, the first waveguide layer and the second waveguide layer are comprised of different materials.

(20) In another example embodiment, a beam delivery system comprises a beam delivery photonic integrated circuit comprising, a plurality of optical inputs, a plurality of waveguide outputs, wherein the plurality of waveguide outputs are tapered, a plurality of beam paths, each beam path connecting one of the plurality of waveguide outputs to only one of the plurality of waveguide inputs, wherein the beam delivery photonic integrated circuit is on a chip, an optical relay assembly, the optical relay assembly configured to receive beams provided by at least two of the plurality of waveguide outputs and focus each received beam on a corresponding plurality of positions of an atomic object confinement apparatus.

(21) In another example embodiment, one or more of the plurality of beam paths comprises an optical element configured to tune a beam for at least one of frequency, phase, or intensity.

(22) In another example embodiment, at least one of the plurality of waveguide outputs is directed to a photodetector.

(23) In another example embodiment, the plurality of beam paths are defined lithographically.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

(1) Having thus described the invention in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

(2) FIG. 1 provides a schematic diagram of an example trapped atomic object quantum computer system, in accordance with an example embodiment.

(3) FIG. 2 provides a schematic diagram of an example controller that may be used in accordance with an example embodiment.

(4) FIG. 3 provides a schematic diagram of an example computing entity that may be used in accordance with an example embodiment.

(5) FIG. 4 provides a block diagram of an example beam delivery photonic integrated circuit, in accordance with an example embodiment.

(6) FIG. 4A provides a schematic diagram of the example beam delivery photonic integrated circuit, in accordance with an example embodiment.

(7) FIG. 5 illustrates an example optical relay assembly, in accordance with an example embodiment.

(8) FIG. 6 provides a flowchart illustrating various processes, procedures, and/or operations performed in fabricating a beam delivery photonic integrated circuit, in accordance with an example embodiment.

(9) FIGS. 7A-7T provide cross-sectional views of various stages of fabricating a beam delivery photonic integrated circuit, in accordance with an example embodiment.

DETAILED DESCRIPTION OF SOME EXAMPLE EMBODIMENTS

(10) The present invention now will be described more fully hereinafter with reference to the

accompanying drawings, in which some, but not all embodiments of the invention are shown. Indeed, the invention may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. The term “or” (also denoted “/”) is used herein in both the alternative and conjunctive sense, unless otherwise indicated. The terms “illustrative” and “exemplary” are used to be examples with no indication of quality level. The terms “generally” and “approximately” refer to within engineering and/or manufacturing limits and/or within user measurement capabilities, unless otherwise indicated. Like numbers refer to like elements throughout.

(11) As described above, in various atomic systems, it is important to be able to precisely and accurately, in terms of position, spacings, mode profile, polarization, frequency, and/or phase, deliver one or more laser beams to the atomic system. For example, in atomic systems such as atomic clocks, Bose-Einstein condensate systems, trapped ion systems, and/or other atomic systems, precise and accurate laser beam delivery is important for various uses of the system, manipulating the system, and/or the like. Various example embodiments corresponding to trapped atomic object (e.g., atoms, ions) quantum computer systems will now be described in more detail.

(12) Exemplary Quantum Computer System

(13) FIG. 1 provides a schematic diagram of an example trapped atomic object quantum computer system **100**, in accordance with an example embodiment. In various embodiments, the quantum computer system **100** comprises a computing entity **10** and a quantum computer **110**. In various embodiments, a controller **30** of the quantum computer **110** may be in communication with the computing entity **10** via one or more wired and/or wireless networks **20**. In various embodiments, the quantum computer **110** comprises the controller **30**, a cryo and/or vacuum chamber **40** enclosing a atomic object confinement apparatus **50** (e.g., an ion trap and/or the like), one or more manipulation sources, one or more laser systems **70**, and/or the like. In various embodiments, the atomic object confinement apparatus **50** is a confinement apparatus configured to atomic object confinement apparatus one or more atomic objects therein and the manipulation sources are configured to provide manipulation signals to one or more portions of the atomic object confinement apparatus **50** via optical paths. In various embodiments, the manipulation signals may be used to initialize one or more atomic objects into a qubit space, perform cooling operations, perform measurement operations, provide one or more gate signals, and/or the like. In various embodiments, the manipulation sources comprise one or more laser systems **70** is configured to provide one or more manipulation signals (e.g., gate signals) to one or more portions of the atomic object confinement apparatus **50** to enact one or more quantum gates (e.g., quantum logic gates). In various embodiments, the quantum gates may be one qubit gates, two qubit gates, and/or the like. In various embodiments, the one or more gate signals may be provided to the one or more portions of the atomic object confinement apparatus **50** via optical path(s) **68**, which may include beam delivery photonic integrated circuit **400** and optical relay assembly **500**. In various embodiments, the atomic objects confined or trapped within the atomic object confinement apparatus **50** are ions, atoms, and/or the like. For example, in an example embodiment, an atomic object is an ytterbium ion. In an example embodiment, an atomic object comprises a qubit ion and a corresponding cooling ion.

(14) In various embodiments, a computing entity **10** is configured to allow a user to provide input to the quantum computer system **100** (e.g., via a user interface of the computing entity **10**) and receive, view, and/or the like output from the quantum computer system **100**. The computing entity **10** may be in communication with the controller **30** via one or more wired or wireless networks **20**. For example, the computing entity **10** may be configured to provide quantum circuits to the controller **30** for execution by the quantum computer **110** and the controller **30** may provide the results of executing one or more quantum circuits to the computing entity **10**.

(15) In various embodiments, the controller **30** is configured to control the atomic object

confinement apparatus **50**, cooling and/or vacuum systems (not shown) controlling the temperature and pressure within the cryo and/or vacuum chamber **40**, manipulation sources, laser system **70**, servo, and/or other components of the quantum computer **110** (e.g., an optical collection system configured for “reading” the output of the quantum computer). In various embodiments, the controller **30** is configured to control various components of the quantum computer **110** in accordance with executable instructions, command sets, and/or the like provided by the computing entity **10** and/or generated by the controller **30**. In various embodiments, the controller **30** is configured to receive output from the quantum computer **110** (e.g., from an optical collection system) and provide the output and/or the result of processing the output to the computing entity **10**.

(16) Exemplary Controller

(17) In various embodiments, a quantum computer **110** comprises a controller **30** configured to control various elements of the quantum computer **110**. In various embodiments, a controller **30** may be configured to cause a quantum computer **110** to perform various operations (e.g., computing operations such as gate operations, cooling operations, transport operations, qubit interaction operations, qubit measurement operations, leakage suppression operations, and/or the like). For example, the controller **30** may be configured to cause manipulation sources to provide manipulation signals to atomic objects confined and/or trapped within the atomic object confinement apparatus **50**. For example, the controller **30** may be configured to cause the laser system **70**, possibly in coordination with servo system, to provide one or more gate signals to one or more atomic objects confined and/or trapped within the atomic object confinement apparatus **50** so as to enact, for example, one or more quantum gates. In various embodiments, the controller **30** may be configured to control a cryogenic system and/or vacuum system controlling the temperature and pressure within the cryogenic and/or vacuum chamber **40**, manipulation sources, and/or other systems controlling the environmental conditions (e.g., temperature, humidity, pressure, and/or the like) within the cryogenic and/or vacuum chamber **40** and/or configured to manipulate and/or cause a controlled evolution of quantum states of one or more atomic objects within the atomic object confinement apparatus **50**.

(18) As shown in FIG. 2, in various embodiments, the controller **30** may comprise various controller elements including processing element(s) **205**, memory **210**, driver controller elements **215**, a communication interface **220**, analog-digital converter **225**, and/or the like. For example, the processing element(s) **205** may comprise programmable logic devices (CPLDs), microprocessors, coprocessing entities, application-specific instruction-set processors (ASIPs), integrated circuits, application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), programmable logic arrays (PLAs), hardware accelerators, other processing devices and/or circuitry, and/or the like, and/or controllers. The term circuitry may refer to an entirely hardware embodiment or a combination of hardware and computer program products. In an example embodiment, the processing element(s) **205** of the controller **30** comprises a clock and/or is in communication with a clock.

(19) For example, the memory **210** may comprise non-transitory memory such as volatile and/or non-volatile memory storage such as one or more of as hard disks, ROM, PROM, EPROM, EEPROM, flash memory, MMCs, SD memory cards, Memory Sticks, CBRAM, PRAM, FeRAM, RRAM, SONOS, racetrack memory, RAM, DRAM, SRAM, FPM DRAM, EDO DRAM, SDRAM, DDR SDRAM, DDR2 SDRAM, DDR3 SDRAM, RDRAM, RIMM, DIMM, SIMM, VRAM, cache memory, register memory, and/or the like. In various embodiments, the memory **210** may store qubit records corresponding the qubits of quantum computer (e.g., in a qubit record data store, qubit record database, qubit record table, and/or the like), a calibration table, an executable queue, computer program code (e.g., in a one or more computer languages, specialized controller language(s), and/or the like), and/or the like. In an example embodiment, execution of at least a portion of the computer program code stored in the memory **210** (e.g., by a processing element(s)

205) causes the controller **30** to perform one or more steps, operations, processes, procedures and/or the like described herein for tracking the phase of an atomic object within an atomic system and causing the adjustment of the phase of one or more manipulation sources and/or signal(s) generated thereby.

(20) In various embodiments, the driver controller elements **215** may include one or more drivers and/or controller elements each configured to control one or more drivers. In various embodiments, the driver controller elements **215** may comprise drivers and/or driver controllers. For example, the driver controllers may be configured to cause one or more corresponding drivers to be operated in accordance with executable instructions, commands, and/or the like scheduled and executed by the controller **30** (e.g., by the processing element(s) **205**). In various embodiments, the driver controller elements **215** may enable the controller **30** to operate a laser system **70**, servo, manipulation sources, vacuum and/or cryogenic systems, and/or the like. In various embodiments, the drivers may be laser drivers; microwave drivers; vacuum component drivers; cryogenic and/or vacuum system component drivers; current drivers, and/or the like. For example, the drivers and/or driver controllers may be configured to cause a magnetic field generation device (e.g., comprising circuitry coupled to a voltage source (e.g., a current driver or voltage driver), permanent magnet(s), and/or a combination thereof) to generate a magnetic field having a particular direction and magnitude at one or more positions of the atomic object confinement apparatus **50**. In various embodiments, a plurality of positions of the atomic object confinement apparatus **50** (e.g., atomic object confinement apparatus zones) may be defined. In various embodiments, the controller **30** comprises means for communicating and/or receiving signals from one or more optical receiver components such as cameras, MEMs cameras, CCD cameras, photodiodes, photomultiplier tubes, and/or the like. For example, the controller **30** may comprise one or more analog-digital converter element(s) **225** configured to receive signals from one or more optical receiver components, calibration sensors, and/or the like.

(21) In various embodiments, the controller **30** may comprise a communication interface **220** for interfacing and/or communicating with a computing entity **10**. For example, the controller **30** may comprise a communication interface **220** for receiving executable instructions, command sets, and/or the like from the computing entity **10** and providing output received from the quantum computer **110** (e.g., from an optical collection system) and/or the result of a processing the output to the computing entity **10**. In various embodiments, the computing entity **10** and the controller **30** may communicate via a direct wired and/or wireless connection and/or one or more wired and/or wireless networks **20**.

(22) Exemplary Computing Entity

(23) FIG. **3** provides an illustrative schematic diagram of an example computing entity **10** that can be used in conjunction with embodiments of the present invention. In various embodiments, a computing entity **10** is configured to allow a user to provide input to the quantum computer **110** (e.g., via a user interface of the computing entity **10**) and receive, display, analyze, and/or the like output from the quantum computer **110**. For example, a user may operate a computing entity **10** to generate and/or program a quantum algorithm and/or quantum circuit (e.g., that includes a D-state AC-Stark shift gate) that may be provided such that the controller **30** may receive the quantum algorithm and/or quantum circuit and cause the quantum computer **110** to perform the quantum algorithm and/or quantum circuit.

(24) As shown in FIG. **3**, a computing entity **10** can include an antenna **312**, a transmitter **304** (e.g., radio), a receiver **306** (e.g., radio), and a processing element **308** that provides signals to and receives signals from the transmitter **304** and receiver **306**, respectively, which may collectively be referred to a transceiver. The signals provided to and received from the transmitter **304** and the receiver **306**, respectively, may include signaling information/data in accordance with an air interface standard of applicable wireless systems to communicate with various entities, such as a controller **30**, other computing entities **10**, and/or the like. In this regard, the computing entity **10**

may be capable of operating with one or more air interface standards, communication protocols, modulation types, and access types. For example, the computing entity **10** may be configured to receive and/or provide communications using a wired data transmission protocol, such as fiber distributed data interface (FDDI), digital subscriber line (DSL), Ethernet, asynchronous transfer mode (ATM), frame relay, data over cable service interface specification (DOCSIS), or any other wired transmission protocol. Similarly, the computing entity **10** may be configured to communicate via wireless external communication networks using any of a variety of protocols, such as general packet radio service (GPRS), Universal Mobile Telecommunications System (UMTS), Code Division Multiple Access 2000 (CDMA2000), CDMA2000 1× (1×RTT), Wideband Code Division Multiple Access (WCDMA), Global System for Mobile Communications (GSM), Enhanced Data rates for GSM Evolution (EDGE), Time Division-Synchronous Code Division Multiple Access (TD-SCDMA), Long Term Evolution (LTE), Evolved Universal Terrestrial Radio Access Network (E-UTRAN), Evolution-Data Optimized (EVDO), High Speed Packet Access (HSPA), High-Speed Downlink Packet Access (HSDPA), IEEE 802.11 (Wi-Fi), Wi-Fi Direct, 802.16 (WiMAX), ultra wideband (UWB), infrared (IR) protocols, near field communication (NFC) protocols, Wibree, Bluetooth protocols, wireless universal serial bus (USB) protocols, and/or any other wireless protocol. The computing entity **10** may use such protocols and standards to communicate using Border Gateway Protocol (BGP), Dynamic Host Configuration Protocol (DHCP), Domain Name System (DNS), File Transfer Protocol (FTP), Hypertext Transfer Protocol (HTTP), HTTP over TLS/SSL/Secure, Internet Message Access Protocol (IMAP), Network Time Protocol (NTP), Simple Mail Transfer Protocol (SMTP), Telnet, Transport Layer Security (TLS), Secure Sockets Layer (SSL), Internet Protocol (IP), Transmission Control Protocol (TCP), User Datagram Protocol (UDP), Datagram Congestion Control Protocol (DCCP), Stream Control Transmission Protocol (SCTP), HyperText Markup Language (HTML), and/or the like.

(25) Via these communication standards and protocols, the computing entity **10** can communicate with various other entities using concepts such as Unstructured Supplementary Service information/data (USSD), Short Message Service (SMS), Multimedia Messaging Service (MMS), Dual-Tone Multi-Frequency Signaling (DTMF), and/or Subscriber Identity Module Dialer (SIM dialer). The computing entity **10** can also download changes, add-ons, and updates, for instance, to its firmware, software (e.g., including executable instructions, applications, program modules), and operating system.

(26) The computing entity **10** may also comprise a user interface device comprising one or more user input/output interfaces (e.g., a display **316** and/or speaker/speaker driver coupled to a processing element(s) **308** and a touch screen, keyboard, mouse, and/or microphone coupled to a processing element(s) **308**). For instance, the user output interface may be configured to provide an application, browser, user interface, interface, dashboard, screen, webpage, page, and/or similar words used herein interchangeably executing on and/or accessible via the computing entity **10** to cause display or audible presentation of information/data and for interaction therewith via one or more user input interfaces. The user input interface can comprise any of a number of devices allowing the computing entity **10** to receive data, such as a keypad **318** (hard or soft), a touch display, voice/speech or motion interfaces, scanners, readers, or other input device. In embodiments including a keypad **318**, the keypad **318** can include (or cause display of) the conventional numeric (0-9) and related keys (#, *), and other keys used for operating the computing entity **10** and may include a full set of alphabetic keys or set of keys that may be activated to provide a full set of alphanumeric keys. In addition to providing input, the user input interface can be used, for example, to activate or deactivate certain functions, such as screen savers and/or sleep modes. Through such inputs the computing entity **10** can collect information/data, user interaction/input, and/or the like.

(27) The computing entity **10** can also include volatile memory or storage **322** and/or non-volatile memory or storage **324**, which can be embedded and/or may be removable. For instance, the non-

volatile memory may be ROM, PROM, EPROM, flash memory, MMCs, SD memory cards, Memory Sticks, CBRAM, PRAM, FeRAM, RRAM, SONOS, racetrack memory, and/or the like. The volatile memory may be RAM, DRAM, SRAM, FPM DRAM, EDO DRAM, SDRAM, DDR SDRAM, DDR2 SDRAM, DDR3 SDRAM, RDRAM, RIMM, DIMM, SIMM, VRAM, cache memory, register memory, and/or the like. The volatile and non-volatile storage or memory can store databases, database instances, database management system entities, data, applications, programs, program modules, scripts, source code, object code, byte code, compiled code, interpreted code, machine code, executable instructions, and/or the like to implement the functions of the computing entity **10**.

(28) Exemplary Laser System and Beam Delivery Photonic Integrated Circuit

(29) A beam delivery photonic integrated circuit **400** may have one or more inputs and has a plurality of outputs. The input(s) and outputs may be connected by waveguides fabricated on a chip, which may be lithographically fabricated on a chip. The beam paths connecting the input(s) and outputs as well as the spacing of the input(s) and outputs may be defined lithographically. The outputs may be a dense set of outputs for output of parallel laser beams. The dense set of outputs may be on a single chip and may allow for the positions of the outputs to be fixed relative to each other, which may provide for greater system stability. A fixed spacing may eliminate the need to make individual optical path adjustments for each beam. The dimensions of the chip, including the input(s) and the outputs may be a few centimeters on each side. The number of outputs may be increased or decreased by adjusting a number of cascading splitters in the beam delivery photonic integrated circuit. The device, however, does not become more complex with the additional beams, which is due in part to the fixed spacing of the outputs. For example, a chip with an edge length of 4 millimeters may provide 100 parallel beams from 100 outputs. With the small size, the chip may be packaged with a shared imaging system of optical relay assembly **500**. Additionally, one or more of the beam delivery photonic integrated circuits may be used together with a shared imaging system to create an array. For example, two beam delivery photonic integrated circuits, each with a 100 beam outputs, could be used together orthogonally to create a 100×100 array. Additionally, the beam delivery photonic integrated circuit **400** may have one or more layers of input(s) and outputs.

(30) As depicted in FIG. **1**, a beam delivery photonic integrated circuit **400** may be located external to the vacuum chamber **40** or, alternatively, a beam delivery photonic integrated circuit may be located inside the vacuum chamber **40**.

(31) In various embodiments, a laser system **70** provides one or more gate signals to one or more portions of an atomic object confinement apparatus **50** having atomic objects confined and/or trapped therein for use as qubits of a quantum computer **110**. For example, if the qubit space is defined based on hyperfine splitting of energy levels of the atomic objects, the frequency of the laser beam may be on the order of 100 MHz to 100 GHz.

(32) A beam delivery photonic integrated circuit **400** may include one or more inputs and two or more outputs. As depicted in FIG. **4**, input(s) and outputs are connected with a plurality of beam paths. Each beam path may include one or more beam splitters, and at least one of the beam paths comprises a modulator. The beam splitters may be configured in a cascading arrangement. For example, the output of a first beam splitter may be connected to the input of another beam splitter. In various embodiments, an input may be connected to an output with a beam path without any splitters. Additionally, or alternatively, a beam delivery photonic integrated circuit **400** may include one input to only one output. Additionally, or alternatively, there may be multiple inputs and outputs each connected by a beam path in the beam delivery photonic integrated circuit **400**.

(33) FIG. **4** provides a block diagram of an example beam delivery photonic integrated circuit **400** in accordance with an example embodiment, and FIG. **4** provides an example of one waveguide layer, which may be in a single plane. The beam delivery photonic integrated circuit **400** may be a photonic integrated circuit fabricated on a chip. The waveguides in one waveguide layer of the beam delivery photonic integrated circuit **400**, as described herein, may be fabricated from one or

more deposited films of waveguide material. The fabrication may use lithography, which may allow for defined geometry of inputs, beam paths, and outputs. The splitters may be made from a single material and configured based on the properties of the material and wavelength of light being used. In various embodiments, there may be multiple waveguide layers such that more than one waveguide layer illustrated in FIG. 4 are in a beam delivery photonic integrated circuit **400**. In an alternative embodiment, there may be a first waveguide layer and a second waveguide layer. In yet another alternative embodiment, there are more than two waveguide layers.

(34) There may be linking of inputs and outputs between layers. For example, an input of a first layer may be an input into a second layer to create a link between the layers. A link between the two layers may be anywhere in the layers, such as, but not limited to, at the input of a stage of splitters or of an optical element or at the output of a splitter or of an optical element. In an alternative embodiment, an input of a first layer may be linked to the input of the second layer such that the output of the second layer is based on, among of things, the inputs to the first and second layers. In an alternative embodiment, an output of a splitter or optical element of a first layer may be linked to the input of a splitter or optical element of a second layer.

(35) In the embodiment depicted by FIG. 4, the output of laser **70** is input to a beam delivery photonic integrated circuit **400** via optical input **410**. In various embodiments, the output of laser **70** may be output on optical fiber and optical input **410** may be a single butt-coupled fiber. In alternative embodiments, the output of laser **70** may be in free-space and input into optical input **410**.

(36) After optical input **410**, beam delivery photonic integrated circuit **400** may split the signal into multiple waveguides using a cascaded series of waveguide-based splitters. FIG. 4 depicts the first splitter as splitter **420**, which has two outputs. Each of the two outputs of splitter **420** goes to another splitter, respectively splitter **430A** and splitter **430B**. Each of the two outputs of splitter **430A** goes to another splitter, respectively splitter **440A** and splitter **440B**. Each of the two outputs of splitter **430B** goes to another splitter, respectively splitter **440C** and splitter **440D**. Thus, as depicted in FIG. 4, there are three stages of splitters that cascade: a first stage of splitter **420**, a second stage of splitters **430A**, **430B**, and a third stage of splitters **440A**, **440B**, **440C**, and **440D**. While FIG. 4 depicts three stages of splitters, other embodiments may have additional stages of cascaded splitters, such as having tens or hundreds of stages of splitters. Additionally, while FIG. 4 depicts splitters in a single plane, the splitters may be configured in a three dimensional direction such that there may be multiple stages of cascading occurring in multiple dimensions (e.g., X, Y, or Z dimensions). Additionally, or alternatively, multiple beam delivery photonic integrated circuits **400** may be provided on one chip, which may be done by isolating in the input and outputs of each beam delivery photonic integrated circuit **400** from each other.

(37) As also depicted in FIG. 4, after the last stage of splitters (depicted as splitters **440A**, **440B**, **440C**, and **440D** in FIG. 4), a beam path may have an optical element. An optical element may allow for a beam in the beam path to be independently tuned, such as for one or more parameters, from beams in other beam paths. In one embodiment, the parameters that may be tuned include frequency, phase, and/or intensity. Additional parameters are discussed herein. In FIG. 4, all but one of the beam paths has an optical element **450** (depicted as **450A**, **450B**, **450C**, **450D**, **450E**, **450F**, and **450G** in FIG. 4), and the one beam path without an integrated circuit element **450** may have a modulator **455** or, alternatively, may not have a modulator **455**. In an example embodiment, N beams from the output of the beam delivery photonic integrated circuit **400** may each be delivered to a particular portion of the atomic object confinement apparatus **50**. For example, a first beam from a first output (e.g., **460A**) and a second beam from a second output (e.g., **460B**) may be delivered to a first portion of the atomic object confinement apparatus **50** and an Nth beam from an Nth output may be delivered to an Nth portion of the atomic object confinement apparatus **50**.

(38) In various embodiments, an optical element **450** may be an interferometer, a phase modulator, photodetector, amplitude modulator, combiner, tapers, polarization control, lens, and/or a

combination of the preceding. For example, the optical element **450** may be configured to provide for control of an output power of a beam. For example, in an example embodiment, wherein an optical element **450** is a Mach-Zehnder interferometer that provides for control of the output power of a corresponding beam, the output power of the corresponding beam may be controlled by resistively heating one arm of the Mach-Zehnder interferometer. For example, the Mach-Zehnder interferometer may act as a phase shifter to power-balance the beams. This may be referred to as thermally tuning the phase of the beam along the path to achieve the desired output power. The output of each of the Mach-Zehnder interferometers may exit the beam delivery photonic integrated circuit **400** through a corresponding output **460**. In an alternative embodiment, the optical element may be a ring modulator. In various embodiments, each of the plurality of outputs **460** of the beam delivery photonic integrated circuit **400** are located and/or disposed along a single polished edge of the beam delivery photonic integrated circuit **400**. Additionally, or alternatively, tapering the waveguide dimensions near an output **460** may allow for and result in a desired output beam profile, such as, for example, a desired mode field or mode profile at the output **460**. The taper of each waveguide output **460** may be the same, as shown in FIG. 4A, or the taper may be different. The beam delivery photonic integrated circuit **400** may be configured so that the stability of the beams' positions and pointing may maintain a constant intensity at the ions at the <0.1% level. Also, the power balance between the focused beams may have equal intensities at all ion zones of the atomic object confinement apparatus **50**, and each beam may have a polarization purity that is 20-dB or better, in various embodiments. The beam delivery photonic integrated circuit **400** may be mechanically polished and finished and may have metal traces to define the Mach-Zehnder interferometer heater elements and their leads. As also depicted in FIG. 4, each beam path may end in an output **460** (depicted as **460A**, **460B**, **460C**, **460D**, **460E**, **460F**, **460G**, and **460H** in FIG. 4). The outputs **460** may be along a single polished edge of the beam delivery photonic integrated circuit **400**. Additionally, in some embodiments, outputs **460** may be located on more than one edge.

(39) In an example embodiment, the beam path that does not have an optical element **450** may be used to provide intensity feedback to the system. For example, the output **460H** of FIG. 4 may be input into a photodetector **80** (e.g., a photodiode) or optical intensity measurement device that may measure the intensity of the beam output via output **460H**. Fast power fluctuations resulting from the input beam coupling variation are common-mode to all beams and can be removed with external feedback based on just one of the output beams, such as with output **460H** of FIG. 4. The modulator **455** may also be a Mach-Zehnder interferometer, in an example embodiment.

(40) Each output **460** may be fixed (relative to the other outputs **460** and/or to the beam delivery photonic integrated circuit **400**) and may provide six degrees of freedom (e.g., freedom in the space of X, Y, and Z coordinates as well as pitch, yaw, and roll). The output may be configured for alignment of an output beam with an ion array and/or with an optical relay assembly and an ion array. The spacing of the outputs **460** may be configured to match needed parameters corresponding to an atomic object confinement apparatus **50**, including, but not limited to, providing a desired mode field and/or mode profile, position, polarization, frequency, phase, focus waist, and/or spacing. For example, the spacing of the outputs **460** may be configured to provide a desired relationship between the mode field diameter of a beam being emitted through an output **460** and the spacing between the output **460** and an adjacent output. Additionally, the degrees of freedom may be provided regardless of how many parallel outputs **460** there may be on a beam delivery photonic integrated circuit **400**. For example, the outputs of a first beam delivery photonic integrated circuit **400** may be aligned with particular positions within a atomic object confinement apparatus **50** based on six degrees of freedom and the outputs of a second beam delivery photonic integrated circuit having twice as many outputs as the first beam delivery photonic integrated circuit may be aligned with particular positions within the atomic object confinement apparatus **50** based on six degrees of freedom. Thus, the alignment of the beam delivery photonic integrated

circuit **400** with the particular positions within an atomic object confinement apparatus **50** does not increase in complexity as the number of outputs **460** of the beam delivery photonic integrated circuit **400** increases, in various embodiments. In one example, an output may be approximately 1 micrometer in diameter and outputs may be spaced approximately 150 micrometers apart.

(41) While FIG. 4 depicts each of the splitters as a 1:2 splitter, in various embodiments a splitter may be a 1:N splitter with a splitter having one input and N outputs. Additionally, or alternatively, the cascading may or may not use splitters having the same ratios (e.g., 1:2, 1:3, 1:4, etc.).

(42) Optical Relay Assembly

(43) Outputs of beam delivery photonic integrated circuit **400** are provided to an optical relay assembly **500**. The optical relay assembly **500** may be configured to provide a desired magnification, such as by configuring a pitch and/or mode-field diameter. As the number of beams from the beam delivery photonic integrated circuit **400** grows, the complexity of the optical relay assembly **500** may or may not increase. For example, the optical relay assembly **500** may be modified to address or correct field-dependent aberrations.

(44) In some embodiments, laser **70** may be a monochromatic laser, which may allow for a single material (e.g., a single glass material) to be used for an optical relay assembly **500**. In other embodiments, more than one material may be used to form or fabricate the optical relay assembly **500**. For example, in an example embodiment wherein the laser **70** is not a monochromatic laser, the optical relay assembly **500** may be formed or fabricated from multiple materials.

(45) FIG. 5 depicts an optical relay assembly **500**. FIG. 5 illustrates an example optical relay assembly **500**. The outputs of beam delivery photonic integrated circuit **400** are input into optical relay assembly **500**. FIG. 5 uses three different colors (e.g., **505A**, **505B**, and **505C**), with each color representing a beam output from one output of beam delivery photonic integrated circuit. The first group of optical elements before the aperture stop may collect light from the photonic chip in a telecentric or non-telecentric fashion, which may depend on the designed output of the chip, and may route it to the second group of optical elements, which may create a telecentric or non-telecentric output where each beam is focused to the required location with the appropriate magnification. FIG. 5 illustrates a plurality of optical elements at **510A-510K** in optical relay assembly **500**. In the illustrated embodiment, optical elements **510A-510K** may be 11 lenses of various shapes, such as concave, convex, biconcave, biconvex, plano-convex, plano-concave, plano-convex, positive meniscus, or negative meniscus. In alternative embodiments, there may be more or less optical elements. The optical elements in the optical relay assembly **500** may be spherical, aspherical, toroidal, or diffractive in nature depending on the designed output of the photonic chip and the required characteristics of the light at the image plane. The optical elements may be anti-reflection coated or uncoated, and may be fabricated out of one or many materials to suit the requirements of the design. The total number of optical elements may vary to meet the design requirements. The system may be composed of all reflective elements, all refractive elements, or a combination of refractive and reflective elements.

(46) The shape of the lenses in optical relay assembly **500** may be configured with a various number of lenses with various shapes in order to deliver beams to the atomic object confinement apparatus precisely and accurately in terms of position, spacings, mode profile, polarization, frequency, and phase.

(47) As depicted in FIG. 5, there may only be one optical relay assembly **500** for a beam delivery photonic integrated circuit **400**. For example, the optical relay assembly **500** may act as a relay assembly for each output **460** of beam delivery photonic integrated circuit **400** that provides a beam to the atomic object confinement apparatus **50**. The beam delivery photonic integrated circuit **400** has outputs that are located in specific positions, which allows for beam outputs of the beam delivery photonic integrated circuit **400** to be known and for an optical relay assembly **500** to be configured based on the position of the outputs. In other embodiments, more than one optical relay assembly **500** may be used with a beam delivery photonic integrated circuit **400**. In other

embodiments, one optical relay assembly **500** may be used with more than one beam delivery photonics integrated circuit **400**. In other embodiments, vacuum chamber **40** may also contain a plurality of optical elements (e.g., **45A**, **45B**).

(48) Fabrication of a Beam Delivery Photonic Integrated Circuit

(49) A beam delivery photonic integrated circuit **400** may be fabricated using photonic circuit fabrication techniques, which may include lithography. The use of lithographically defined geometry may allow for tighter tolerances, which may lead to greater stability in the quantum computer. A waveguide layer in a beam delivery photonic integrated circuit may be etched to form the desired waveguide geometry, including each of the splitters in the beam delivery photonic integrated circuit **400**. With the process of FIG. **6**, or variations of it, beam delivery photonic integrated circuit **400** may be fabricated on a single chip. Although the device fabrication described herein utilizes only one waveguide layer, integrated circuits can be made multi-level, including with splitters, couplers and/or waveguides enabling the delivery of multiple beam arrays with different wavelengths.

(50) In various embodiments, a waveguide layer may be achromatic and may facilitate multiple wavelengths of light, such as broadband infrared light to ultraviolet light. In embodiments that use ultraviolet and visible lasers, which may be required for addressing atomic objects confined and/or trapped in an atomic object confinement apparatus, the device may be fabricated using specialty materials optimized for ultraviolet or visible light wavelengths. In some embodiments a waveguide layer may use a single mode waveguide. In other embodiments, a multimode waveguide layer may be used. In some embodiment, a first layer may be made of a first material and a second layer may be made of a second material.

(51) FIG. **6** provides a flowchart illustrating various processes, procedures, and/or operations performed in fabricating a beam delivery photonic integrated circuit **400**. FIGS. **7A-7T** provide cross-sectional views of various stages of fabricating a beam delivery photonic integrated circuit **400**.

(52) At step **602** of FIG. **6**, a first cladding layer **704** is deposited on the substrate **702** (e.g., Si), as shown in FIGS. **7A** and **7B**. For example, a first cladding layer **704** comprising oxide and/or dielectric (e.g., SiO₂) may be deposited on the substrate **702**. In various embodiments, the first cladding layer **704** may electrically and/or thermally isolate the waveguide layer **706** from the substrate **702**. FIG. **7A** illustrates a cross-section before completion of step/operation **602**, and FIG. **7B** illustrates a cross-section after completion of step **602**.

(53) Continuing with FIG. **6**, at step **604**, a waveguide layer **706** is deposited onto the first cladding layer **704**, as shown in FIG. **7C**. The waveguide layer **706** may be thin film that is deposited. For example, a waveguide layer **706** comprising the material (e.g., a dielectric material, such as Al₂O₃, Si₃N₄, HfO₂, AlN, Ta₂O₅, and/or the like) of the passive waveguide layer **706** may be deposited onto the first cladding layer **704**. FIG. **7C** illustrates a cross-section after completion of step **604**.

(54) Returning to FIG. **6**, at step **606**, a hard mask layer **708** is deposited onto the waveguide layer **706**. The hard mask layer may be of the same material as the first cladding layer **704** (e.g., SiO₂), or it may be of a different material. For example, a hard mask layer **708** is deposited onto the waveguide layer **706**. For example, FIG. **7D** illustrates a cross-section after completion of step/operation **606**.

(55) Continuing with FIG. **6**, at step **608**, a resist layer **710**, such as a photoresist, is added by spinning. For example, one or more drops of resist layer **710** material may be deposited on the surface of the hard mask layer **708** and the substrate **702** and layers deposited and/or bonded thereon may be spun at a high rate so that the one or more drops of resist layer **710** material become evenly dispersed across the surface of the hard mask layer **708**. FIG. **7E** illustrates a cross-section after completion of step **608**. For example, a resist layer **710** is deposited onto the hard mask layer **708**.

(56) Returning to FIG. 6, at step **610**, the resist layer **710** may be baked and then exposed to UV radiation to develop resist layer **710A**. FIG. 7F illustrates a cross-section after completion of step **610**.

(57) Continuing with FIG. 6, at step **612**, the hard mask layer **708** is etched to form hard mask layer **708A**. In various embodiments, a photolithography and/or mask etching process may be used to etch the hard mask layer to form hard mask layer **708A**. FIG. 7G illustrates a cross-section after completion of step **612**.

(58) Returning to FIG. 6, at step **614**, the resist layer **710** is removed. FIG. 7H illustrates a cross-section after completion of step/operation **614**.

(59) Continuing with FIG. 6, at step **616**, the waveguide layer **706** is etched and/or patterned to form waveguide layer **706A**. FIG. 7I illustrates a cross-section after completion of step **616**.

(60) Continuing with FIG. 6, at step **618**, a second cladding layer is deposited to form cladding layer **708B**. FIG. 7J illustrates a cross-section after completion of step **618**. Step **618** may, or may not, result in cladding being deposited such that the exposed surface at **708B** requires smoothing or flattening. If the exposed surface is to be smooth or flattened, an optional polishing step may occur at step **620**. For example, the exposed surface of the cladding layer **708B** may be polished using chemical mechanical polishing (CMP) such that the exposed surface is smooth and/or flat to form cladding layer **7C**. FIG. 7K illustrates a cross-section after completion of step **620**. If the optional step **620** does not occur due to the exposed surface not needing to be smoothed or flattened, the process continues step **622** shown in FIG. 6.

(61) At step **622**, a metal layer may be deposited to form metal layer **712**. For example, the metal layer may be made of a material (Al, Ti, Au, and/or the like), and the metal layer may define a Mach Zehnder interferometer heater element and/or its lead. FIG. 7L illustrates a cross-section after completion of step **622**.

(62) Continuing with FIG. 6, at step **624**, a resist layer **714**, such as a photoresist, is added by spinning FIG. 7M illustrates a cross-section after completion of step **624**.

(63) Returning to FIG. 6, at step **626**, the resist layer **714** is baked and exposed to UV radiation and to develop resist layer **714A**. FIG. 7N illustrates a cross-section after completion of step **626**.

(64) Continuing with FIG. 6, at step **628**, the metal layer **712** is etched to form metal layer **712A**. For example, the metal layer **712** may be etched form a metal layer **712A** that defines one or more Mach Zehnder interferometer heater elements, corresponding leads, and/or the like. FIG. 7O illustrates a cross-section after completion of step **628**.

(65) Returning to FIG. 6, at step **630**, the resist layer **714A** is removed. FIG. 7P illustrates a cross-section after completion of step/operation **630**.

(66) Continuing with FIG. 6, at step **632**, a resist layer **716**, such as a photoresist, is added by spinning FIG. 7Q illustrates a cross-section after completion of step **632**.

(67) At step **634** shown in FIG. 6, the resist layer **716** is baked and exposed to UV radiation and to develop resist layer **716A**. FIG. 7R illustrates a cross-section after completion of step **626**.

(68) Returning to FIG. 6, at step **636**, cladding layer **708B** is etched to form cladding layer **708C** and cladding layer **704** is etched to form cladding layer **704A**. FIG. 7S illustrates a cross-section after completion of step **636**.

(69) Continuing with FIG. 6, at step **638**, the resist layer **716A** is cleaned and removed. FIG. 7T illustrates a cross-section of a beam delivery photonic integrated circuit **400** after completion of step **638**.

(70) After cleaning, an edge with the outputs of the beam delivery photonic integrated circuit **400** may be polished.

Technical Advantages

(71) Various embodiments provide technical solutions to the technical problem of precisely and accurately, in terms of frequency, frequency offset, and/or phase, delivering multiple laser beams (e.g., ultraviolet laser beams, visible laser beams, infrared laser beams, etc.) in a dense array (e.g.,

to a dense array of atomic object confinement apparatus zones). In an embodiment, the array of atomic object confinement apparatus zones may be a one-or-two-dimensional array of zones within an atomic object confinement apparatus. The beam delivery photonic integrated circuit may be configured to deliver the laser beams so that they may be mutually parallel and correctly spaced to overlap with the atomic object confinement apparatus zones with submicron accuracy. Further, lithographically defined geometry of the waveguides provide greater stability. Additionally, a beam delivery photonic integrated circuit may be configured to provide more than one beam while being sized to fit in a mezzanine breadboard surrounding a vacuum chamber. A beam delivery photonic integrated circuit **400** on a single chip may also be easier to use in a quantum computing system compared to other waveguides that comprise multiple components (e.g., fiber optics, troughs holding fiber optics, couplers, splitters, interferometers, modulators, etc., particularly when such components may occupy a large amount of space. Additionally, a beam delivery photonic integrated circuit **400** on a single chip allows the beam delivery photonic integrated circuit **400** to be placed closed to the vacuum chamber **40**. Moreover, the alignment of the plurality of beams provided by a beam delivery photonic integrated circuit **400** is simpler than traditional means and does not increase in complexity and/or in the number of degrees of freedom with an increase in the number of outputs of the beam delivery photonic integrated circuit **400**. In particular, various embodiments provide an improvement over traditional means, which may be collections of individual fibers, collimators, and mirrors for each beam, but these systems required a large footprint on the mezzanine breadboard. Further, these systems were limited in the degree to which they may scale by the amount of available space and ability to package the current laser-beam delivery apparatus.

(72) Various embodiments of the beam delivery photonic integrated circuit **400** provide for a system allowing for delivery of a plurality of laser beams that may be power balanced and have a constant intensity at the $<0.1\%$ level. Embodiments therefore provide an improvement to the operation of a quantum computer.

(73) Various embodiments also allow for tuning at once with six degrees of freedom without requiring the tuning of each individual beam, which allows for an increasing number of outputs without increasing complexity of alignment.

CONCLUSION

(74) Many modifications and other embodiments of the invention set forth herein will come to mind to one skilled in the art to which the invention pertains having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the invention is not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A beam delivery system comprising: a beam delivery photonic integrated circuit comprising: one or more optical inputs; a plurality of waveguide outputs; and a plurality of beam paths, each beam path connecting one of the plurality of waveguide outputs to at least one of the one or more optical inputs, wherein the plurality of waveguide outputs are configured to emit a plurality of parallel beams, and wherein the beam delivery photonic integrated circuit is on a chip; and a telecentric optical relay assembly, the telecentric optical relay assembly configured to receive the plurality of parallel beams provided by the plurality of waveguide outputs and focus each received beam on a corresponding one of a plurality of positions of an atomic object confinement apparatus in a telecentric manner.

2. The beam delivery system of claim 1, wherein one or more beam paths of the plurality of beam

paths comprises a respective optical element configured to tune a respective beam for at least one of frequency, phase, or intensity.

3. The beam delivery system of claim 2, wherein the respective optical element is one of an interferometer, a phase modulator, a photodetector, an amplitude modulator, a beam combiner, a beam splitter, a lens, a polarization controller, or a combination of two or more thereof.

4. The beam delivery system of claim 1, wherein the atomic object confinement apparatus is disposed within a vacuum chamber and the beam delivery system comprises one or more optical elements disposed within the vacuum chamber.

5. The beam delivery system of claim 4, wherein the one or more optical elements disposed within the vacuum chamber comprise the beam delivery photonic integrated circuit.

6. The beam delivery system of claim 1, wherein the plurality of beam paths are defined lithographically.

7. The beam delivery system of claim 1, wherein the plurality of waveguide outputs are located on a polished edge of the beam delivery photonic integrated circuit.

8. The beam delivery system of claim 1, wherein at least one of the one or more optical inputs is optically connected to two or more of the plurality of waveguide outputs via respective beam paths of the plurality of beam paths, the respective beam paths comprising one or more beam splitters.

9. The beam delivery system of claim 1, wherein each of the plurality of waveguide outputs are at most 150 micrometers from a next closest of the plurality of waveguide outputs.

10. The beam delivery system of claim 1, wherein an optical input of the one or more optical inputs is butt-coupled to an optical fiber providing an input beam.

11. The beam delivery system of claim 1, wherein the beam delivery photonic integrated circuit comprises a first waveguide layer and a second waveguide layer, a first portion of the plurality of beam paths disposed at least in part in the first waveguide layer and a second portion of the plurality of beam paths disposed at least in part in the second waveguide layer.

12. The beam delivery system of claim 11, wherein at least one of the first waveguide layer or the second waveguide layer comprises a dielectric material.
